

A
Project Review-IV Presentation
on
ChatPCCOE: A Web-Based Communication and Collaboration Platform for PCCOE Staff and Students

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Platform Overview



Centralized Digital Ecosystem

PCCOE Connect integrates **students**, **faculty**, and **administrators** into a unified digital environment, enabling seamless communication, academic resource sharing, and institutional activity management through a single, secure platform.

The system transforms traditional academic communication by providing **real-time collaboration tools**, **secure authentication**, and **scalable infrastructure** designed specifically for engineering institutions.

Challenges in Traditional Systems

Fragmented Communication

Lack of a centralized platform leads to scattered information across multiple channels and systems

Resource Sharing Difficulties

Academic materials and updates distributed through disconnected methods reduce accessibility

Scalability Concerns

Managing large user bases exceeding 5,000 students requires robust, efficient infrastructure

Security Vulnerabilities

Authentication challenges and data protection risks in traditional academic systems

Project Objectives



Enhanced Communication

Enable seamless collaboration between all stakeholders in the academic ecosystem



Secure Authentication

Implement enterprise-grade security protocols for user verification and data protection



Data Management

Centralize academic information for efficient organization and retrieval



System Scalability

Support growing user bases with responsive, cloud-based infrastructure



User Experience

Provide intuitive interface accessible across all devices and platforms



Resource Sharing

Facilitate instant updates and academic resource distribution

Technology Stack

Frontend

- React.js with TypeScript
- Tailwind CSS framework
- Responsive UI components

Backend

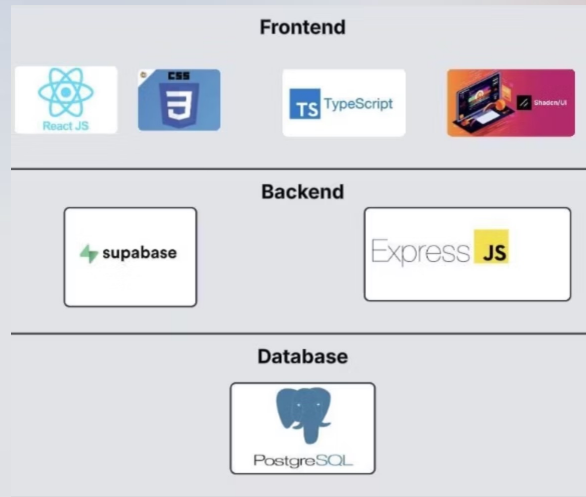
- Node.js runtime
- Express.js framework
- RESTful API endpoints

Database

- Supabase platform
- PostgreSQL database
- Real-time data sync

DevOps

- Docker containerization
- Kubernetes orchestration
- Vercel and AWS deployment
- CI/CD automation



Literature Survey

Research on Related Technologies

Survey of recent research papers and projects relevant to web-based academic systems and related technology stacks.

Title	Objective	Technology	Published Date
NEXA Learning Management System	To build a modular learning platform for managing courses, attendance, and academic activities	MongoDB, Express.js, React.js, Node.js	2024
E-Learning Web Application Using MERN Stack	To develop an online learning platform that allows students to access study materials and courses digitally	MongoDB, Express.js, React.js, Node.js	2024
MERN Stack Powered Academic Management Solution	To develop a system for managing attendance, marks, notices, and communication between students and faculty	MongoDB, Express.js, React.js, Node.js	2024
Academic Log Portal – Student Management System	To provide a centralized portal for academic records, attendance tracking, and institutional communication	Web Application Technologies	2024
MERN Stack Based Online Learning Platform	To provide a digital platform for tutors and students to manage courses and online learning resources	MongoDB, Express.js, React.js, Node.js	2025
Student Management System: A Web-Based Solution for Academic Administration	To streamline academic activities such as courses, attendance, and feedback using a centralized web system	Web Technologies, Database Systems	2025
College Management System (CMS)	To automate administrative and academic operations within colleges using a web-based system	Full-Stack Web Development	2025

System Architecture

PCCOE Connect System Architecture

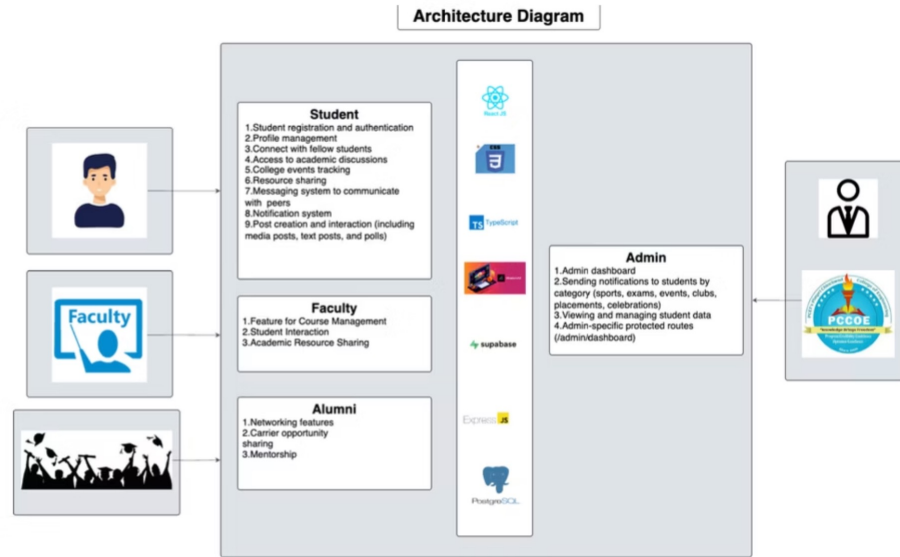


Figure 1: PCCOE Connect System Architecture

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Security & Authentication Architecture

Authentication Flow

- OAuth 2.0 integration with Auth0
- JWT token generation and validation
- Role-based access control (RBAC)
- Session management with refresh tokens
- Multi-factor authentication support

Security Measures

- End-to-end encryption for messages
- HTTPS/TLS for all communications
- Password hashing with bcrypt
- SQL injection prevention
- CORS policy enforcement
- Rate limiting on API endpoints
- Regular security audits

Content Moderation Architecture

Our platform integrates a robust backend moderation service to automatically screen user-generated posts and comments for inappropriate content before publication, ensuring a safe and respectful digital environment.

Architecture Flow



Backend Implementation

- Implemented using **Supabase Edge Functions**
- **JWT authentication** for user validation
- Built-in **profanity/bad-word list**
- **Regex-based** content checking (case-insensitive)
- **JSON response:** { flagged: boolean, reason: string }

Result

- **Clean Content:** Post/comment created successfully
- **Flagged Content:** Submission blocked and error message shown to user

Types of Testing & Test Cases

Types of Testing

Unit Testing



- Tested individual modules (login, messaging, resource upload).
- Each function works independently.

Integration Testing



- Frontend (React) & backend (Supabase) communication.
- Smooth API and data flow.

System Testing



- Complete system including authentication, messaging, notifications, resource sharing.
- Overall functionality validated.

Test Cases Implemented

Login Authentication



- Valid login → Success | Invalid login → Error
- Unauthorized → Blocked

Messaging System



- Send → Real-time delivery | Receive → Instantly visible
- Multiple users → No delay

Resource Sharing



- Upload → Stored successfully | Download → Accessible
- File validation → Correct formats only

Validation & Testing Outcome

Validation and Quality Assurance



Performance Testing

- Fast loading and smooth UI
- **Real-time** updates using Supabase
- Handles multiple users efficiently



Security Testing

- **Secure** authentication using Supabase Auth
- Data encryption via HTTPS
- Role-based access control

Outcome of Testing

- All features working correctly
- System is **stable, secure, and scalable**
- Improved user experience with smooth performance
- Reliable **real-time** communication

Implementation and Results



Authentication System

Secure user verification with Auth0 and JWT tokens



Messaging Platform

Real-time communication between users



REST API

Endpoints for data exchange and integration



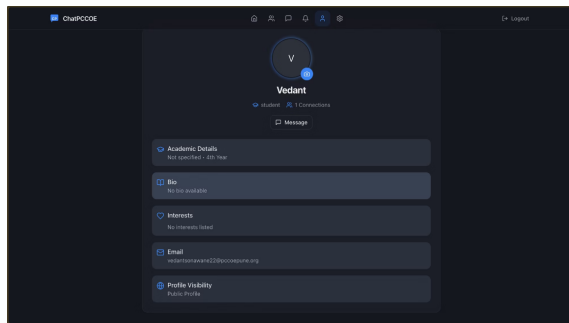
Dashboard Interface

Unified portal for all academic activities



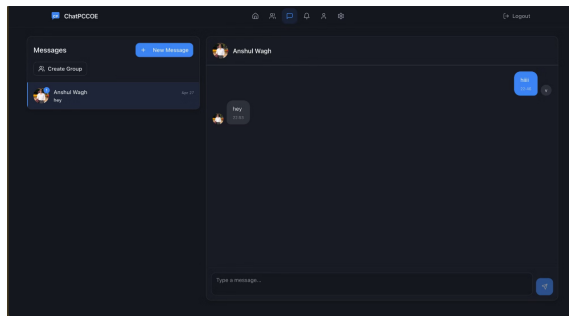
Notifications

Instant alerts for updates and announcements



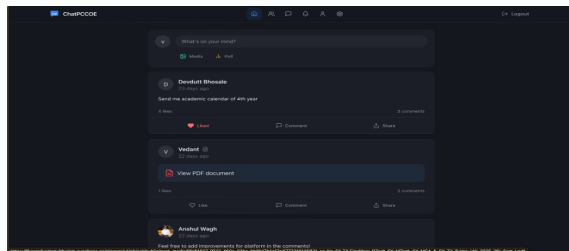
User Profile Page

- Shows **personal and academic details** such as year, email, and profile visibility settings.
- Allows users to **customize their profile** with bio, interests, and profile picture for better identity representation.



Messaging System

- Enables **real-time one-to-one communication** between users with a clean chat interface.
- Includes features like **new message creation and group chats**, improving collaboration among students.



Home Feed (Posts Page)

- Displays a **centralized feed** where students can post updates, share PDFs, and interact through likes, comments, and shares.
- Provides **engagement features** like media upload, polls, and real-time interaction to enhance student communication.

Advantages and Applications

Key Advantages



Highly Scalable Architecture

Cloud-based infrastructure supports growing user bases efficiently



Enterprise Security

Auth0 and JWT ensure data protection and authentication



Real-Time Features

Instant notifications and live updates across platform

Institutional Applications

- **Student Portal**

Course materials, assignments, and academic updates

- **Faculty Management**

Resource sharing and collaborative teaching tools

- **Event Coordination**

Announcements, workshops, and campus activities

- **Placement Support**

Training updates and career services integration

- **Library Services**

Digital resources and academic content distribution

The platform combines scalable infrastructure, enterprise-grade security, and real-time capabilities to support diverse institutional needs with clear, readable interfaces tuned to the Creme palette for consistent visual hierarchy.

Conclusion and Future Scope



Conclusion

PCCOE Connect delivers a scalable, secure, and efficient platform for academic communication and collaboration, transforming institutional operations with modern cloud infrastructure and user-centric design.

Future Enhancements

AI-Powered Intelligence

Smart recommendations for resources and personalized learning paths

Mobile Application

Native iOS and Android apps for on-the-go access

Advanced Analytics

Data dashboards for institutional performance insights

Enhanced Collaboration

Video conferencing and interactive virtual classrooms